

Margin Requirements

Simplified Rule-Based Framework

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Risk Modelling

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1. Overview

This document specifies a simplified, rule-based margin framework for margin trading products. Margin requirements are expressed as fixed percentage rates per asset class, applied directly as minimum equity conditions on the outstanding loan. No econometric modelling is required.

The framework rests on a single unifying principle: the **haircut is the margin requirement**. An initial margin haircut h_{IM} of, say, 30% means the customer must hold equity equal to at least 30% of the current market value at all times during the IM phase. Equivalently, the outstanding loan must not exceed $(1 - h_{IM})$ times the market value. This identity connects the haircut directly to the LTV monitoring limit.

The framework consists of four margin components:

1. **Initial Margin (IM)** — minimum equity condition at position open, enforced for one trading day (the IM horizon). Expressed as rate h_{IM} on market value.
2. **Maintenance Margin (MM)** — ongoing minimum equity condition from day two onwards. Expressed as rate $h_{MM} \leq h_{IM}$ on market value, giving the customer more headroom once the initial risk window has passed.
3. **Overnight Margin** — additional equity cushion required at end-of-day (EOD) to cover the overnight risk window. Expressed as add-on rate h_{ON} applied on top of the prevailing phase rate.
4. **Mark-to-Market (MTM) Margin** — continuous revaluation of the outstanding loan against current market prices. Updates the LTV in real time; a breach of the applicable LTV limit triggers immediate auto-liquidation.

2. Core Definitions

2.1 Market Value and Loan

Let N denote the number of units held in instrument i and P_t the current market price. The **market value** at time t is:

$$MV_t = N \times P_t$$

The **outstanding loan** L_t is the principal borrowed plus accrued interest to date. Customer equity is:

$$E_t = MV_t - L_t$$

2.2 Loan-to-Value Ratio

The **LTV ratio** measures the loan relative to the current market value:

$$LTV_t = \frac{L_t}{MV_t}$$

An LTV of zero means the position is fully equity-funded. An LTV of one means the customer has no equity remaining. Because margin trading limits leverage, the LTV is always strictly below one.

2.3 The Haircut as LTV Limit

The core identity of this framework is that **the haircut h defines both the margin requirement and the LTV monitoring limit**:

$$\begin{aligned} \text{Margin requirement: } E_t &\geq h \times MV_t \\ \text{Equivalent LTV condition: } LTV_t &= \frac{L_t}{MV_t} \leq 1 - h \end{aligned}$$

The LTV limit for a given phase is therefore $1 - h$, where h is the applicable margin rate. A higher margin rate produces a lower (more conservative) LTV limit.

Symbol	Description
h_{IM}	Initial margin rate; applies on day of opening ($t = 0$)
h_{MM}	Maintenance margin rate; applies from $t = 1$ onwards
h_{ON}	Overnight margin add-on; applied at EOD on top of prevailing phase rate
T_{IM}	IM horizon: 1 trading day
L_t	Outstanding loan at time t
MV_t	Market value at time t
LTV_t	Loan-to-Value ratio at time t

3. Margin Components

3.1 Initial Margin (IM)

Initial Margin applies at the moment a position is opened and remains in force for one trading day ($T_{IM} = 1$). It is the most conservative condition, ensuring the customer enters the position with sufficient equity to absorb an adverse intraday move.

$$\text{IM condition: } LTV_0 = \frac{L_0}{MV_0} \leq 1 - h_{IM}$$

The maximum loan at opening and the minimum required equity follow directly:

$$L_{\max} = (1 - h_{IM}) \times MV_0 \quad E_{\min} = h_{IM} \times MV_0$$

The auto-liquidation price during the IM phase (the price at which $LTV = 1 - h_{IM}$ exactly) is:

$$P_{IM}^* = \frac{L_0}{N \times (1 - h_{IM})}$$

3.2 Maintenance Margin (MM)

After one trading day, the position transitions from the IM phase to the Maintenance Margin phase. The MM rate $h_{MM} \leq h_{IM}$ is less conservative, raising the LTV limit and giving the customer more room. The same structural condition applies:

$$\text{MM condition: } LTV_t = \frac{L_t}{MV_t} \leq 1 - h_{MM} \quad (t \geq 1)$$

The auto-liquidation price in the MM phase:

$$P_{MM}^* = \frac{L_t}{N \times (1 - h_{MM})}$$

Because $h_{MM} < h_{IM}$, we have $1 - h_{MM} > 1 - h_{IM}$ and therefore $P_{MM}^* < P_{IM}^*$: the position can absorb a larger adverse price move before liquidation fires in the MM phase.

3.3 Overnight Margin

At end-of-day, an additional overnight add-on h_{ON} is applied on top of the prevailing phase rate. This reflects the extended risk window during which continuous intraday monitoring is unavailable.

$$\text{Overnight condition: } LTV_t^{ON} = \frac{L_t}{MV_t} \leq 1 - h_{\text{phase}} - h_{ON}$$

where h_{phase} is h_{IM} on day 0 and h_{MM} from day 1 onwards. The overnight LTV limit is therefore tighter (lower) than the intraday limit by exactly h_{ON} .

If the overnight condition is not met at EOD, the customer must either:

1. deposit additional cash to reduce the LTV below $1 - h_{\text{phase}} - h_{ON}$, or
2. partially close the position before the EOD cut-off.

Failure to remedy the breach triggers automatic liquidation of the excess exposure.

The overnight auto-liquidation price:

$$P_{ON}^* = \frac{L_t}{N \times (1 - h_{\text{phase}} - h_{ON})}$$

3.4 Mark-to-Market (MTM) Margin

MTM reflects the continuous revaluation of the open position as market prices move. It is not a separate cash margin call but a real-time update to the LTV:

$$\begin{aligned} \text{MTM}_t &= (P_t - P_{t-1}) \times N \\ \text{LTV}_t &= \frac{L_t}{N \times P_t} \end{aligned}$$

A negative MTM (adverse price move) raises the LTV; a positive MTM lowers it. The LTV is recomputed **continuously** at each price update throughout the trading day. As soon as LTV_t breaches the applicable limit, auto-liquidation is triggered immediately — there is no grace period.

Auto-liquidation rule: at any time t , if $LTV_t > 1 - h_{\text{phase}}$ (intraday) or $LTV_t > 1 - h_{\text{phase}} - h_{\text{ON}}$ (overnight check), the position is automatically liquidated at market prices. No margin call is issued.

4. Two-Phase Monitoring Framework

4.1 Phase Transition

The IM horizon is exactly one trading day. The transition from IM to MM is automatic:

$$h_{\text{phase}}(t) = \begin{cases} h_{\text{IM}} & t = 0 \text{ (day of opening)} \\ h_{\text{MM}} & t \geq 1 \end{cases}$$

4.2 Intraday Monitoring (Both Phases)

At every price update the LTV is recomputed and checked against the intraday limit:

$$LTV_t = \frac{L_t}{N \times P_t} \leq 1 - h_{\text{phase}}(t)$$

4.3 EOD Overnight Check (Both Phases)

At end-of-day, the overnight add-on tightens the limit:

$$LTV_t^{\text{ON}} = \frac{L_t}{N \times P_t^{\text{close}}} \leq 1 - h_{\text{phase}}(t) - h_{\text{ON}}$$

4.4 LTV Limits Summary

Phase	Time	LTV limit	Action on breach
IM — intraday	$t = 0$	$1 - h_{\text{IM}}$	Auto-liquidation
IM — overnight	$t = 0$ EOD	$1 - h_{\text{IM}} - h_{\text{ON}}$	Top-up / reduce or auto-liquidation
MM — intraday	$t \geq 1$	$1 - h_{\text{MM}}$	Auto-liquidation
MM — overnight	$t \geq 1$ EOD	$1 - h_{\text{MM}} - h_{\text{ON}}$	Top-up / reduce or auto-liquidation

5. Asset Class Rate Table

Fixed margin rates are maintained per asset class. All rates are expressed as a percentage of current market value.

Asset Class	h_{IM}	h_{MM}	h_{ON}
Major index equities (e.g. EuroStoxx 50, S&P 500 constituents)	15%	10%	3%
Other equities	30%	25%	5%
Equity ETFs — broad market	15%	10%	3%
Equity ETFs — sector / thematic	20%	15%	4%
Investment-grade bonds ($\geq$ BBB)	10%	7%	2%
High-yield bonds ($<$ BBB)	30%	20%	5%

Rates are reviewed at least quarterly. Individual instruments may be assigned higher rates at the discretion of Risk if recent volatility or liquidity conditions warrant it.

6. Worked Example

6.1 Setup

A customer opens a leveraged long position in an “Other equity” (non-index stock). Applicable rates: $h_{\text{IM}} = 30\%$, $h_{\text{MM}} = 25\%$, $h_{\text{ON}} = 5\%$.

Parameter	Value
Current price P_0	€100.00 per share
Number of shares N	150
Market value MV_0	$150 \times €100 = €15,000$
Customer equity E_0	€5,000
Outstanding loan L_0	$€15,000 - €5,000 = €10,000$
Leverage	$3\times$

Applicable LTV limits:

Phase	Rate(s)	LTV limit	Liquidation price P^*
IM intraday	$h_{IM} = 30\%$	$1 - 0.30 = 70\%$	$€10,000 / (150 \times 0.70) = €95.24$
IM overnight	$h_{IM} + h_{ON} = 35\%$	$1 - 0.35 = 65\%$	$€10,000 / (150 \times 0.65) = €102.56$
MM intraday	$h_{MM} = 25\%$	$1 - 0.25 = 75\%$	$€10,000 / (150 \times 0.75) = €88.89$
MM overnight	$h_{MM} + h_{ON} = 30\%$	$1 - 0.30 = 70\%$	$€10,000 / (150 \times 0.70) = €95.24$

6.2 Step 1 — Opening Check (IM Phase, $t = 0$)

$$LTV_0 = \frac{€10,000}{€15,000} = 66.7\% \leq 70\% \Rightarrow \text{Position may be opened.}$$

Minimum equity the customer must post:

$$E_{\min} = h_{IM} \times MV_0 = 30\% \times €15,000 = €4,500$$

Customer posted €5,000 > €4,500: condition satisfied with a buffer of €500.

The IM intraday auto-liquidation price:

$$P_{IM}^* = \frac{L_0}{N \times (1 - h_{IM})} = \frac{€10,000}{150 \times 0.70} = €95.24$$

The stock must fall –4.76% from €100 before intraday auto-liquidation triggers.

6.3 Step 2 — EOD Overnight Check ($t = 0$ EOD)

The overnight LTV limit tightens to $1 - h_{IM} - h_{ON} = 65\%$. At EOD price €100 (unchanged intraday):

$$LTV_0^{ON} = \frac{€10,000}{€15,000} = 66.7\% > 65\% \Rightarrow \text{Overnight condition breached.}$$

The overnight auto-liquidation price:

$$P_{ON}^* = \frac{€10,000}{150 \times 0.65} = €102.56$$

At the current price of €100 the LTV already exceeds the 65% overnight limit. The customer must post additional collateral before EOD cut-off.

Required top-up to clear the overnight condition exactly:

$$L_{\max}^{ON} = 65\% \times €15,000 = €9,750 \Rightarrow \text{Cash to post} = L_0 - L_{\max}^{ON} = €10,000 - €9,750 = €250$$

After posting €250: $L = €9,750$, $LTV^{ON} = 65\% = \text{limit}$. Passed.

6.4 Step 3 — Transition to MM Phase ($t = 1$)

After one trading day the IM phase ends and MM governs. The LTV limit widens from 70% to 75% (intraday). Assuming price remains €100 and loan is €9,750 after the overnight top-up:

$$LTV_1 = \frac{€9,750}{€15,000} = 65.0\% \leq 75\% \Rightarrow \text{MM phase check passed.}$$

The new MM intraday auto-liquidation price:

$$P_{MM}^* = \frac{€9,750}{150 \times 0.75} = €86.67$$

The position now tolerates a –13.3% price decline before auto-liquidation — significantly more room than the IM phase.

6.5 Step 4 — Intraday MTM: Price Falls –5% to €95

MTM is calculated continuously. At price €95:

$$\text{MTM} = (95 - 100) \times 150 = -\text{€}750 \quad \text{MV} = 150 \times \text{€}95 = \text{€}14,250$$

$$\text{LTV} = \frac{\text{€}9,750}{\text{€}14,250} = 68.4\% \leq 75\% \Rightarrow \text{Still open.}$$

6.6 Step 5 — Intraday MTM: Price Falls Further to €88

$$\text{MV} = 150 \times \text{€}88 = \text{€}13,200$$

$$\text{LTV} = \frac{\text{€}9,750}{\text{€}13,200} = 73.9\% \leq 75\% \Rightarrow \text{Still open.}$$

6.7 Step 6 — Intraday MTM: Price Falls to €86.50

$$\text{MV} = 150 \times \text{€}86.50 = \text{€}12,975$$

$$\text{LTV} = \frac{\text{€}9,750}{\text{€}12,975} = 75.1\% > 75\% \Rightarrow \text{AUTO-LIQUIDATION TRIGGERED.}$$

The position is automatically liquidated at market price €86.50. The customer receives the residual equity:

$$E_{\text{final}} = \text{€}12,975 - \text{€}9,750 = \text{€}3,225$$

The customer has lost $\text{€}5,000 - \text{€}3,225 = \text{€}1,775$ (35.5% of their initial equity) due to the –13.5% price decline.

6.8 Summary Table

Event	Price	Loan	MV	LTV	Limit / Status
Open ($t = 0$, IM intraday)	€100	€10,000	€15,000	66.7%	$\leq 70\%$ Open
EOD overnight ($t = 0$)	€100	€10,000	€15,000	66.7%	$\leq 65\%$ Breach
After top-up €250	€100	€9,750	€15,000	65.0%	$\leq 65\%$ Cleared
MM phase open ($t = 1$)	€100	€9,750	€15,000	65.0%	$\leq 75\%$ Open
MTM –5% ($t = 1$)	€95	€9,750	€14,250	68.4%	$\leq 75\%$ Open
MTM –12% ($t = 1$)	€88	€9,750	€13,200	73.9%	$\leq 75\%$ Open
MTM –13.5% ($t = 1$)	€86.50	€9,750	€12,975	75.1%	$> 75\%$ LIQUIDATE

7. Comparison with Alternative Margin Frameworks

The table below applies all four frameworks to the same position: **150 shares at €100, market value €15,000, customer equity €5,000, outstanding loan €10,000 (3× leverage)**. Where a framework does not permit the full 3× position, the maximum feasible position on €5,000 equity is shown instead.

	IB — Reg T	IB — Portfolio Margin	TR Model-Based	TR Simplified (this doc)
Core methodology	Flat regulatory rule (50% of MV, Reg T / FINRA Rule 4210)	OCC scenario grid $\pm 15\%$ across 31 price points (TIMS); 15% single-stock floor	Filtered historical simulation + EWMA volatility scaling; 99% RVaR over 5-day horizon	Fixed $h_{IM}/h_{MM}/h_{ON}$ per asset class; haircut = LTV limit ($LTV \leq 1 - h$)
Initial margin	€7,500 (50% of MV)	€2,250 (15% of MV)	€1,710 (11.4% of MV)	$E_{min} = h_{IM} \times MV = 30\% \times €15,000 = €4,500$; $LTV_0 = 66.7\% \leq 70\%$
3× leverage possible?	No. Requires €7,500 equity; max 2× on €5,000	Yes. €5,000 > €2,250	Yes. €5,000 $\gg$ €1,710	Yes. $LTV_0 = 66.7\% \leq 70\% = 1 - h_{IM}$
IM horizon	Until position is closed (50% applies always)	Until position is closed (PM grid applies always)	4 trading days, then MM regime	1 trading day , then MM regime
Maintenance / ongoing check	25% of current MV; Excess Liquidity ≥ 0	15% scenario floor; Excess Liquidity ≥ 0	LTV monitored at 75% / 80% / 90% thresholds (graduated)	$LTV \leq 1 - h_{MM} = 75\%$; single hard limit
Overnight treatment	Overnight reverts to Reg T 50% if intraday was lower	PM grid re-evaluated; same floor applies overnight	EOD haircut add-on raises LTV toward breach thresholds	LTV limit tightens to $1 - h_{MM} - h_{ON} = 70\%$ at EOD
LTV monitoring principle	Equity / MV $\geq 25\%$ (maintenance)	NLV – PM_req ≥ 0	Graduated: monitoring 75%, breach 80%, liquidation 90%	Single limit: $LTV \leq 1 - h$; no graduated warnings
Liquidation trigger	Excess Liquidity < 0 (binary)	Excess Liquidity < 0 (binary)	LTV > 90% (after warnings at 75% and 80%)	$LTV > 1 - h_{phase}$ (immediate, no prior warning)
Intraday liquidation price P^*	€5,000 / (100 × 0.75) = €66.67 (on adjusted 2× position)	€10,000 / (150 × 0.85) = €78.43	€10,000 / (150 × 0.90 × 0.90) = €82.30	IM phase: €10,000 / (150 × 0.70) = €95.24 MM phase: €10,000 / (150 × 0.75) = €88.89
Price fall to intraday liquidation	–33.3% (large buffer; 2× position)	–21.6%	–17.7%	IM phase: –4.8% MM phase: –11.1%
MTM frequency	Continuous (Excess Liquidity re-checked at each mark)	Continuous	Continuous (LTV re-computed per price update)	Continuous ($LTV = L/MV_t$ at each price update)
Asset-class differentiation	Partial (penny stocks, leveraged ETFs get higher rates; standard = flat 50%)	Partial (single-stock 15% vs. 10% for broad indices)	Full (per-ISIN vol model; h_{floor} table override)	Full (per-class h_{IM} , h_{MM} , h_{ON} table)
Model complexity	None (rule-based)	Low (fixed scenario grid, OCC-licensed)	High (EWMA, filtered hist. simulation, correlation break, stress scenarios)	Low (table lookup; one LTV computation per price tick)

Notes. IB Reg T uses the adjusted 2× position (100 shares, €10,000 MV) because 3× cannot be opened. All other frameworks use 150 shares, €15,000 MV, €10,000 loan. TR Simplified uses $h_{IM} = 30\%$, $h_{MM} = 25\%$, $h_{ON} = 5\%$ (other equity class). TR Model-Based liquidation price uses $h_{eff} = 10\%$ and the 90% LTV liquidation threshold. Highlighted rows show the liquidation price and price fall — the primary practical differentiators.